

## Review Questions 1-17

### 1. Explain the differences in resource and data forks used in macOS.

In older versions of macOS, a file consists of two parts: the data fork and the resource fork. The data fork contains the data the user creates, such as text or spreadsheets. Applications also read and write to it. The resource fork stores additional information like menus, dialog boxes, and other application-related metadata. Together they make up a complete Mac file.

### 2. Main challenges in acquiring an image of a system running macOS? (Choose all that apply)

**a, c, and d.** Most commercial forensics software doesn't fully support macOS, macOS is incompatible with most write-blockers, and you need special tools (and sometimes Apple factory training) to remove drives or open cases, particularly with Mac Minis and MacBook Airs.

### 3. To recover a password in macOS, which tool do you use?

**c. Keychain Access.** Keychain Access is the Mac application used to manage and restore passwords for applications, websites, and other system files.

### 4. What are the major improvements in the Linux Ext4 file system?

Ext4 added support for partitions larger than 16 TB, improved management of large files, and offered more flexibility compared to Ext3. It is now considered the standard file system for most Linux distributions.

### 5. How does macOS reduce file fragmentation?

macOS uses clumps, which are groups of contiguous allocation blocks. By allocating files in these contiguous groups rather than scattered individual blocks, fragmentation is reduced.

### 6. Linux is the only OS that has a kernel. True or False?

**False.** All UNIX-like operating systems have a kernel, and so do all Windows operating systems. A kernel is not unique to Linux.

### 7. Hard links work in only one partition or volume. True or False?

**True.** Hard links point to a specific inode, and inodes are unique within a single partition or volume. They cannot span across different partitions or volumes (unlike symbolic links, which can point to items on other drives or network locations).

**8. Which Linux system file contains hashed passwords?**

**d. /etc/shadow.** This is where hashed passwords for local user accounts are stored on Linux systems.

**9. Which describes the superblock's function? (Choose all that apply)**

**b and c.** The superblock specifies the disk geometry and available space (b) and manages the file system, including configuration information (c). The boot block (not the superblock) stores bootstrap code, so (a) is incorrect. Inodes handle links to data, not the superblock, so (d) is incorrect.

**10. What's the disk arbitration feature used for in macOS?**

Disk arbitration is the mount function in macOS. Certain forensic tools (like BlackBag Technologies Macintosh Forensic Software and SubRosaSoft MacForensicsLab) can disable disk arbitration, which turns off the automatic mount function. This allows you to connect a suspect drive to a Mac without needing a separate write-blocking device, preventing the OS from altering evidence on the drive.

**11. In Linux, which is the home directory for the superuser?**

**b. root.** The superuser's home directory is /root (not /home/superuser or any other option).

**12. Which certifies when an OS meets UNIX requirements?**

**c. The Open Group.** The Open Group is the organization that certifies whether an operating system meets UNIX requirements.

**13. On most Linux systems, current user login information is in which location?**

**d. /var/log/utmp.** This file stores current user login information on Linux systems.

**14. Hard links are associated with which of the following?**

**b. A specific inode.** A hard link is a pointer that allows accessing the same file by different filenames, and all hard links to a file point to the same inode.

**15. Which describes plist files? (Choose all that apply)**

**c. They're preference files for applications.** Plist files store preference settings for installed applications on macOS systems. They are not limited to Linux (they're a macOS feature), they don't necessarily require a special editor, and they don't require special installers.

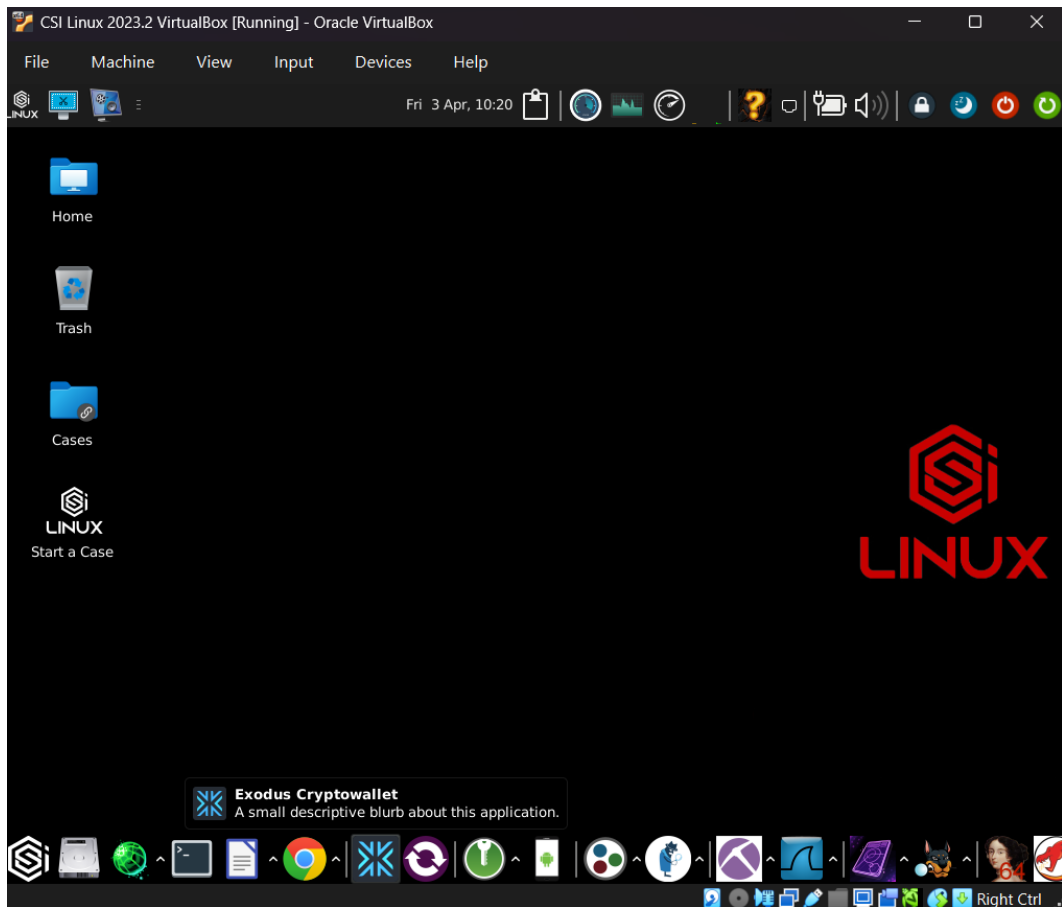
**16. Data blocks contain actual files and directories and are linked directly to inodes. True or False?**

**True.** Data blocks are where directories and files are stored on a disk drive, and their locations are linked directly to inodes.

**17. Which is a new file added in macOS? (Choose all that apply)**

**a and c.** /private/var/db (a) and /var/db/diagnostics (c) are new file locations added in macOS related to unified logging and diagnostics.

**CSI Linux**



**CSI Linux 2023.2 Virtual...**  
Running

### Display

Video Memory: 32 MB  
Graphics Controller: VMSVGA  
Remote Desktop Server: Disabled  
Recording: Disabled

### Storage

Controller: IDE  
IDE Secondary Device 0: [Optical Drive] Empty  
Controller: LsiLogic  
SCSI Port 0: CSI Linux 2023.2-disk001.vmdk (Normal, 50.00 GB)

### Audio

Host Driver: Default  
Controller: ICH AC97

### Network

Adapter 1: Intel PRO/1000 MT Desktop (NAT Network, 'Public Network')  
Adapter 2: Intel PRO/1000 MT Desktop (Internal Network, 'Whonix')